Prime Factors and Integer Exponents

Concept

Prime factors reveal shared structure. Integer exponents include zero and negatives.

GCF / LCM

Use smaller shared prime powers / larger powers of all primes.

Negative power

$a^{-n} = 1/a^n$ for $a \neq 0$; it does not mean a negative answer.

Laws

Add exponents in products; subtract in quotients; multiply in powers.

Scientific notation

$c \times 10^n$, with $1 \leq |c| < 10$.

Example 1

Find GCF and LCM of 72 and 120.

Solution

1. $72 = 2^3 \times 3^2$; $120 = 2^3 \times 3 \times 5$.
2. GCF $= 2^3 \times 3 = 24$.
3. LCM $= 2^3 \times 3^2 \times 5 = 360$.

Prime Factors and Integer Exponents

Example 2

Simplify $(3x^{-2}y^{3})/(6xy^{-1})$.

Solution

1. Original restrictions: $x \neq 0$, $y \neq 0$.

2. Combine: $(1/2)x^{-3}y^{4} = y^{4}/(2x^{3})$.

3. Keep both original restrictions.

Prime Factors and Integer Exponents

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find GCF and LCM of 84 and 126.

2. Simplify 5^{-2}.

3. Simplify $(4x^3)/(2x^{-2})$, including restrictions.

4. Write 0.000072 in scientific notation.

5. Evaluate $(-2)^{-3}$.

Prime Factors and Integer Exponents

Practice answers | check after trying

1. $84 = 2^2 \times 3 \times 7$; $126 = 2 \times 3^2 \times 7$. GCF $= 42$ and LCM $= 252$.

2. $1/25$.

3. $2x^5$, $x \neq 0$.

4. 7.2×10^{-5}.

5. $1/(-2)^3 = -1/8$.

Multiplying Polynomials

Concept

Multiply every term in one polynomial by every term in the other.

Distribute

$(a + b)(c + d) = ac + ad + bc + bd$.

Square

$(a \pm b)^2 = a^2 \pm 2ab + b^2$.

Difference of squares

$(a + b)(a - b) = a^2 - b^2$.

Check

Count products, preserve signs, then collect like terms.

Example 1

Expand $(2x - 3)(x^2 + x + 5)$.

Solution

1. Distribute: $2x^3 + 2x^2 + 10x - 3x^2 - 3x - 15$.

2. Collect terms: $2x^3 - x^2 + 7x - 15$.

Multiplying Polynomials

Example 2

Expand $(3x - 2)^2$.

Solution

1. Use $a^2 - 2ab + b^2$ with $a = 3x$, $b = 2$.

2. Result: $9x^2 - 12x + 4$.

Multiplying Polynomials

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Expand $(x + 6)(x + 2)$.

2. Expand $(2x + 1)(3x - 4)$.

3. Expand $(x - 5)^2$.

4. Expand $(4x + 3)(4x - 3)$.

5. Expand $(x + 2)(x^2 - x + 3)$.

Multiplying Polynomials

Practice answers | check after trying

1. $x^2 + 8x + 12$.

2. $6x^2 - 8x + 3x - 4 = 6x^2 - 5x - 4$.

3. $x^2 - 10x + 25$.

4. $16x^2 - 9$.

5. $x^3 - x^2 + 3x + 2x^2 - 2x + 6 = x^3 + x^2 + x + 6$.

Factoring Polynomials

Concept

Factoring reverses expansion: write a sum as a product.

First step

Take out the greatest common factor.

Trinomial

For $x^2 + bx + c$, find $p + q = b$ and $pq = c$.

Square difference

$a^2 - b^2 = (a - b)(a + b)$.

Finish

Factor completely; expand to check.

Example 1

Factor $x^2 - x - 20$.

Solution

1. Find numbers with product -20 and sum -1: -5 and 4.

2. Result: $(x - 5)(x + 4)$.

Factoring Polynomials

Example 2

Factor $3x^3 - 27x$ completely.

Solution

1. Take out $3x$: $3x(x^2 - 9)$.

2. Factor the square difference: $3x(x - 3)(x + 3)$.

Factoring Polynomials

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Factor $8x^2 - 12x$.

2. Factor $x^2 + x - 30$.

3. Factor $x^2 - 10x + 25$.

4. Factor $4x^2 - 49$.

5. Explain why $x^2 + 9$ is not $(x + 3)(x - 3)$.

Factoring Polynomials

Practice answers | check after trying

1. $4x(2x - 3)$.

2. $(x + 6)(x - 5)$.

3. $(x - 5)^2$.

4. $(2x - 7)(2x + 7)$.

5. The proposed product expands to $x^2 - 9$; its constant has the wrong sign.

Relations and Functions

Concept

A function gives exactly one output for each allowed input.

Domain / range

Allowed inputs / resulting outputs.

Vertical-line test

A function has at most one intersection with any vertical line.

Notation

$f(4)$ asks for an output; $f(x) = 4$ asks which inputs give 4.

Context

Use discrete sets for counts; intervals for continuous inputs.

Example 1

Is $\{(1,4), (2,4), (3,7)\}$ a function? State its domain and range.

Solution

1. Yes: each input has only one output. Repeated outputs are allowed.

2. Domain $\{1,2,3\}$; range $\{4,7\}$.

Relations and Functions

Example 2

For $f(x) = 3x - 4$, find $f(-2)$ and solve $f(x) = 8$.

Solution

1. Evaluate: $f(-2) = 3(-2) - 4 = -10$.

2. Solve: $3x - 4 = 8 \rightarrow x = 4$.

Relations and Functions

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Is $x = y^2$ a function of x over the full real relation? Explain.

2. For $g(x) = -2x + 5$, find $g(3)$ and solve $g(x) = 9$.

3. State domain and range for $\{(0,-1), (2,3), (4,3)\}$.

4. Explain why a continuous line may misrepresent a ticket-sales model.

Relations and Functions

Practice answers | check after trying

1. No. At $x = 4$, both $y = 2$ and $y = -2$ occur.

2. $g(3) = -1$; $-2x + 5 = 9$ gives $x = -2$.

3. Domain $\{0,2,4\}$; range $\{-1,3\}$.

4. It suggests fractional tickets are allowed, so separate points are appropriate when only whole tickets can be sold.

Slope and Equations of Lines

Concept

Slope measures rate of change. A line can be written in several equivalent forms.

Slope

$m = (y_2 - y_1)/(x_2 - x_1)$, for different x-values.

Forms

$y = mx + b$; $y - y_1 = m(x - x_1)$; $Ax + By + C = 0$.

Parallel / perpendicular

Equal slopes / slope product -1, when defined.

Special lines

Horizontal: $y = b$, slope 0. Vertical: $x = a$, slope undefined.

Example 1

Find the line through (2,3) and (6,11).

Solution

1. $m = (11 - 3)/(6 - 2) = 2$.

2. $y - 3 = 2(x - 2) \rightarrow y = 2x - 1$.

Slope and Equations of Lines

Example 2

Find the line through $(2,5)$ perpendicular to $y = (2/3)x - 1$.

Solution

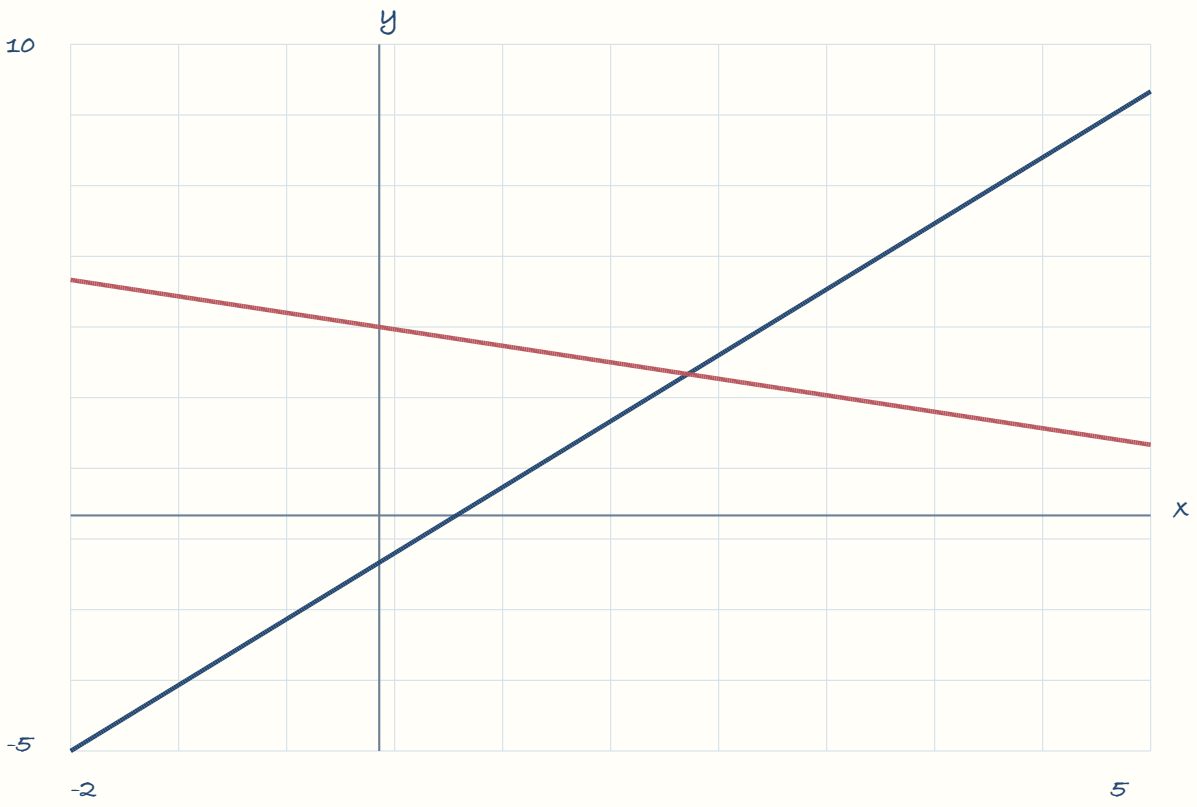
1. Perpendicular slope $= -3/2$.

2. $y - 5 = (-3/2)(x - 2) \rightarrow y = (-3/2)x + 8$.

Slope and Equations of Lines

Visual notes | read the graph

Blue: $y = 2x - 1$; red: $y = -x/2 + 4$



The slopes multiply to −1. These lines are perpendicular, even if unequal axis scales alter the visual angle.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Slope and Equations of Lines

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find slope through (-1,4) and (3,-8).

2. Write a line through (2,7) with slope 3.

3. Find the line perpendicular to $y = -x/4 + 2$ through the origin.

4. Write the vertical line through (5,-2).

5. Are $y = 2x + 1$ and $4x - 2y + 2 = 0$ distinct parallel lines?

Slope and Equations of Lines

Practice answers | check after trying

1. $(-8 - 4)/(3 + 1) = -3$.

2. $y - 7 = 3(x - 2)$, so $y = 3x + 1$.

3. $y = 4x$.

4. $x = 5$.

5. No. The second simplifies to $y = 2x + 1$; they are the same line, with infinitely many common points.

Arithmetic Sequences

Concept

An arithmetic sequence has a constant difference between consecutive terms.

General term

$a_n = a_1 + (n - 1)d$; n is a positive integer.

Known terms

$d = (a_q - a_p)/(q - p)$.

Membership

A target is a term only if its index n is allowed.

Optional sum

$S_n = n(a_1 + a_n)/2$.

Example 1

Given $a_4 = 18$ and $a_{10} = 42$, find the general term.

Solution

1. $d = (42 - 18)/(10 - 4) = 4$.

2. $a_1 = 18 - 3(4) = 6$.

3. $a_n = 6 + 4(n - 1) = 4n + 2$.

Arithmetic Sequences

Example 2

Find term 40 of 12, 17, 22, ...

Solution

1. $a_1 = 12$ and $d = 5$.

2. $a_{40} = 12 + 39(5) = 207$.

Arithmetic Sequences

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find d and a_{20} for $30, 27, 24, \ldots$.

2. If $a_3 = 11$ and $a_8 = 31$, find a_1.

3. Which term of $5, 12, 19, \ldots$ equals 103?

4. Optional: find the sum of the first 12 positive multiples of 3.

Arithmetic Sequences

Practice answers | check after trying

1. $d = -3$; $a_{20} = 30 + 19(-3) = -27$.

2. $d = (31 - 11)/5 = 4$; $a_1 = 11 - 8 = 3$.

3. $103 = 5 + 7(n - 1)$, so $98 = 7(n - 1)$, giving $n = 15$.

4. First 3, last 36, count 12: $S = 12(3 + 36)/2 = 234$.

Systems of Linear Equations

Concept

A system solution satisfies both equations at once: it is their graph intersection.

Substitution

Replace an isolated variable in the other equation.

Elimination

Make coefficients opposite, then add the equations.

Number of solutions

Intersecting: one; parallel: none; same line: infinitely many.

Check

Substitute the ordered pair into both original equations.

Example 1

Solve $y = 2x + 1$ and $x + y = 10$.

Solution

1. Substitute: $x + (2x + 1) = 10$.
2. $3x = 9 \rightarrow x = 3$; then $y = 7$.
3. Solution $(3,7)$; it satisfies both equations.

Systems of Linear Equations

Example 2

Solve $2x + 3y = 13$ and $3x - 3y = 12$.

Solution

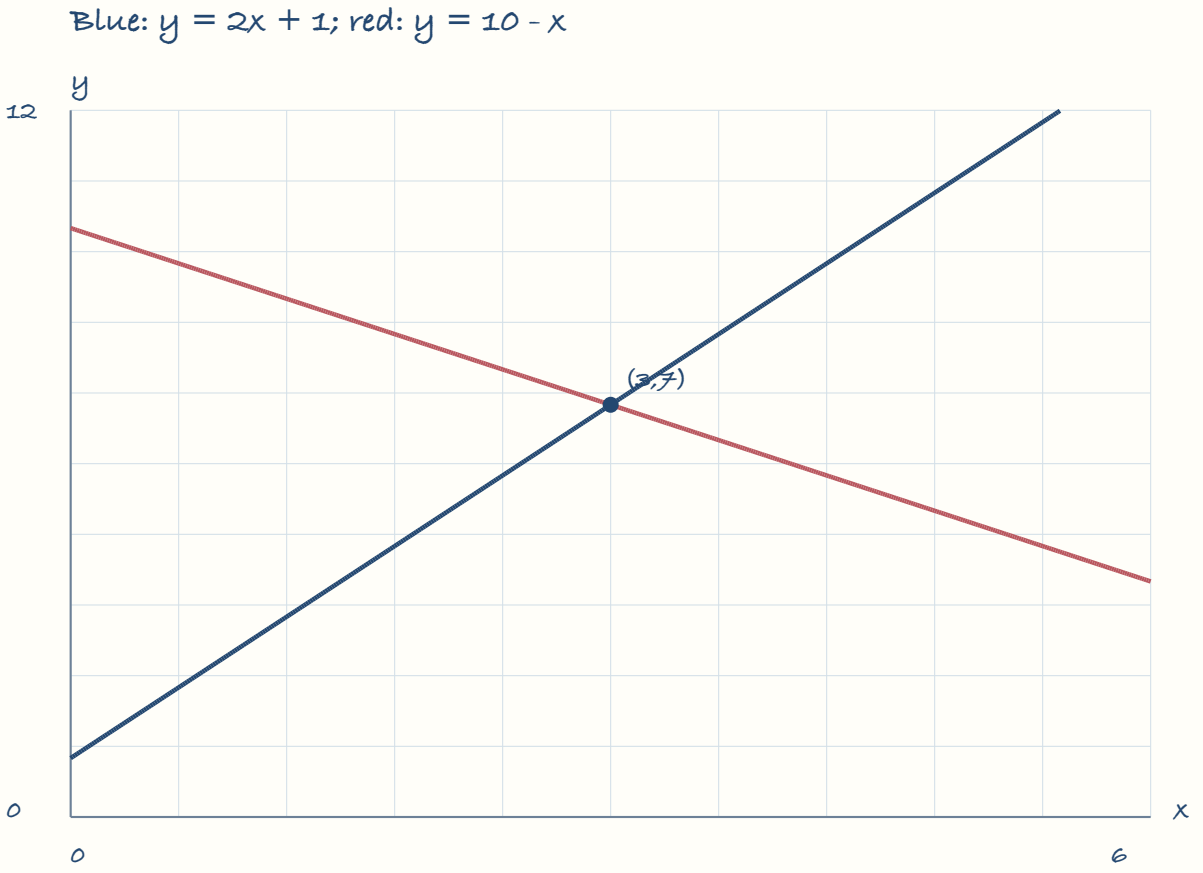
1. Add equations: $5x = 25 \rightarrow x = 5$.

2. Substitute: $10 + 3y = 13 \rightarrow y = 1$.

3. Solution $(5,1)$.

Systems of Linear Equations

Visual notes | read the graph



The intersection satisfies both equations. Substitution confirms the graph reading.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Systems of Linear Equations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Solve $y = x - 2$ and $x + y = 8$.

2. Solve $3x + 2y = 16$ and $x - 2y = 0$.

3. Classify $x + y = 4$ and $2x + 2y = 8$.

4. Two plans cost $A = 10 + 2v$ and $B = 4v$. Find their break-even use and cost.

Systems of Linear Equations

Practice answers | check after trying

1. $2x - 2 = 8$ gives $x = 5$, $y = 3$.

2. Adding gives $4x = 16$, so $x = 4$ and $y = 2$.

3. Infinitely many solutions: the equations describe the same line.

4. $10 + 2v = 4v$ gives $v = 5$ visits; both cost $20. Below 5 visits B is cheaper; above 5 A is cheaper.

Right-Triangle Trigonometry

Concept

Trigonometric ratios connect an acute angle with side lengths in a right triangle.

SOH CAH TOA

$\sin\theta = o/h$; $\cos\theta = a/h$; $\tan\theta = o/a$.

Label first

Hypotenuse is opposite 90°; o and a depend on the chosen angle.

Find an angle

Use inverse trig and degree mode for this unit.

Applications

Elevation/depression use a horizontal reference; include eye height.

Example 1

Find the opposite side when $\theta = 35°$ and the hypotenuse is 10 cm.

Solution

1. $\sin 35° = o/10$.

2. $o = 10\sin 35° \approx 5.74$ cm.

Right-Triangle Trigonometry

Example 2

A ramp rises 1.2 m over a horizontal run of 8 m. Find its angle.

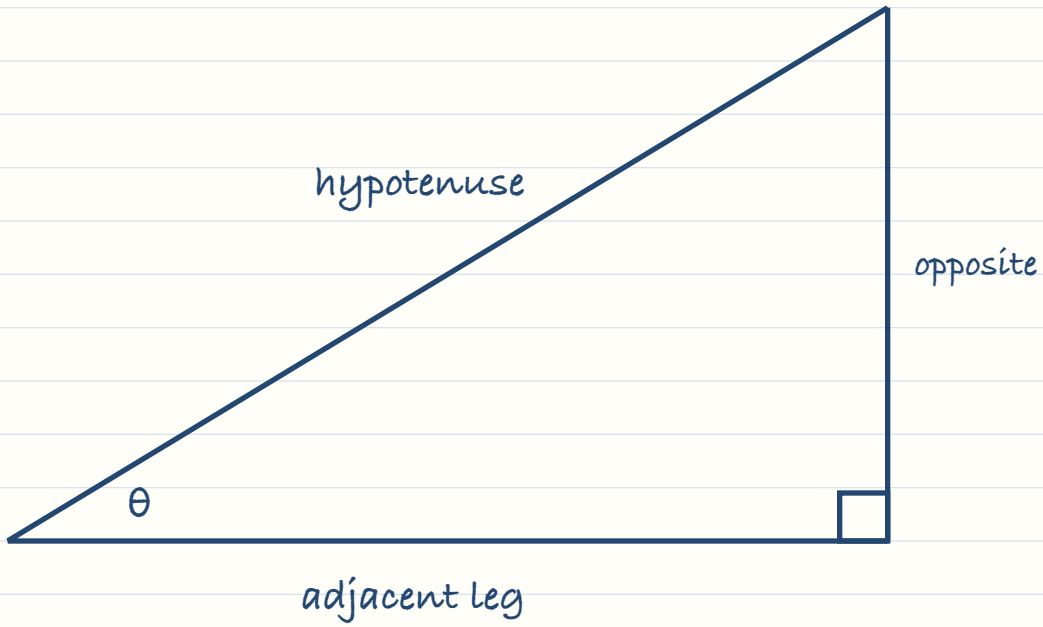
Solution

1. $\tan\theta = 1.2/8 = 0.15$.

2. $\theta = \tan^{-1}(0.15) \approx 8.53°$.

Right-Triangle Trigonometry

Visual notes | label before calculating



The hypotenuse is opposite the right angle. Opposite and adjacent refer to the selected acute angle θ. A different selected angle changes those two labels.

Pythagoras: $a^2 + b^2 = c^2$. For right triangles, sin θ = opposite/hypotenuse, cos θ = adjacent/hypotenuse, and tan θ = opposite/adjacent.

Right-Triangle Trigonometry

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find the opposite side for $\theta = 30°$ and hypotenuse 12.

2. Find θ when opposite $= 7$ and adjacent $= 9$.

3. Find hypotenuse when adjacent $= 5$ and $\theta = 60°$.

4. Why must you use horizontal distance, not the slope length, as adjacent in an elevation problem?

Right-Triangle Trigonometry

Practice answers | check after trying

1. $12 \sin 30^\circ = 6$.

2. $\theta = \tan^{-1}(7/9) \approx 37.87^\circ$.

3. $h = 5/\cos 60^\circ = 10$.

4. The adjacent leg lies along the horizontal reference line; the slope length is the hypotenuse, so using it as adjacent changes the triangle ratio.

Gross Pay, Deductions and Net Pay

Concept

Gross pay is before deductions. Net pay is what remains after deductions.

Earnings

Hourly rate × hours; commission rate × sales; add base pay if given.

Net pay

Gross pay − total deductions.

Periods

Usually 26 biweekly or 24 semi-monthly payments in a stated annual model.

Tax brackets

Each rate applies only to its bracket portion. Practice rates are hypothetical.

Example 1

Earn $18/h for 40 regular hours and 5 overtime hours at 1.5 times the rate.

Solution

1. Regular = $40(18) = \$720$; overtime rate = $18(1.5) = \$27/h$.

2. Gross = $720 + 5(27) = \$855$.

Gross Pay, Deductions and Net Pay

Example 2

A hypothetical tax is 10% on the first $20,000 and 20% above that. Income: $30,000.

Solution

1. Tax $= 0.10(20{,}000) + 0.20(10{,}000) = \$4000$.

2. Average rate $= 4000/30{,}000 \approx 13.33\%$; marginal rate $= 20\%$.

Gross Pay, Deductions and Net Pay

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find gross pay for 32 h at $21/h.

2. Find 3% commission on $12,000 of sales plus a $400 base.

3. Gross pay is $1400. Stated deductions are 12% of gross plus $45. Find net pay.

4. Under the hypothetical brackets above, find tax and average rate on $25,000.

Gross Pay, Deductions and Net Pay

Practice answers | check after trying

1. $672.

2. $0.03(12{,}000) + 400 = \$760$.

3. Percent deduction = $168; total deductions = $213; net = $1187.

4. Tax $= 2000 + 0.20(5000) = \$3000$. Average rate $= 3000/25{,}000 = 12\%$. Do not apply 20% to the entire income.